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Chapter 1

Lecture 1 - 06/10/2026

1.1 Fundamentals of Stellar Hydrodynamics: One-Dimensional Models

To fully understand the essential meaning of the equations, analyze the physical consequences and have an intuitive understanding of the situation, we will work primarily in **one dimension**.

From a hydrodynamic point of view, the treatment is based on two fundamental descriptions that will be used alternately depending on convenience:

- The **Eulerian description**
- The **Lagrangian description**

1.1.1 Eulerian Description

In the **Eulerian description**, position and time are chosen as independent variables. Operating generally in spherical coordinates, position is defined by the distance from the center of the sphere. We define the independent variables:

- r: distance from the center (with $r \in [0, R]$, where R is the total radius of the star).
- t: time.

All other physical quantities will be referred to these two independent variables. For example, in the general case, the plasma density will depend on the distance from the center and on time: $\rho = \rho(r, t)$. Similarly, the mass enclosed within a sphere of radius r will be a function $m = m(r, t)$.

Derivation of the Continuity Equation

Let us consider a mass element dm, which is the mass contained in an infinitesimal spherical shell of thickness dr. The area of this shell is $4\pi r^2$. Multiplying the area by the thickness dr gives the volume, and multiplying the latter by the density ρ yields the mass contained in the shell under static conditions.

However, in a dynamic situation, it is necessary to also consider the mass flux entering and leaving the shell. The matter flux through the surface of the

shell is given by the product of the density ρ, the plasma velocity v and the time interval dt.

Therefore, the mass variation can be described by analyzing the partial derivatives. Taking the partial derivative of mass with respect to r, we consider only the static contribution (since the flux term does not depend solely on the spatial geometry at a fixed time):

$$\frac{\partial m}{\partial r} = 4\pi r^2 \rho \tag{1.1}$$

Differentiating this expression now with respect to time t, we obtain the mixed second derivative:

$$\frac{\partial^2 m}{\partial t \partial r} = 4\pi r^2 \frac{\partial \rho}{\partial t} \tag{1.2}$$

Let us now perform the reverse procedure. Taking the partial derivative of the mass with respect to time t, we obtain the expression of the mass flux through the surface:

$$\frac{\partial m}{\partial t} = -4\pi r^2 \rho v \tag{1.3}$$

Differentiating this quantity with respect to the radius r, we apply the product rule of differentiation:

$$\frac{\partial^2 m}{\partial r \partial t} = -8\pi r \rho v - 4\pi r^2 \frac{\partial(\rho v)}{\partial r} \tag{1.4}$$

By Schwarz's theorem on mixed derivatives, the two derived expressions must be equal. Equating them and bringing all terms to the same side, we obtain:

$$4\pi r^2 \frac{\partial \rho}{\partial t} + 8\pi r \rho v + 4\pi r^2 \frac{\partial(\rho v)}{\partial r} = 0 \tag{1.5}$$

Dividing the entire equation by $4\pi r^2$, we isolate the time derivative of the density:

$$\frac{\partial \rho}{\partial t} = -\frac{2}{r}\rho v - \frac{\partial(\rho v)}{\partial r} \tag{1.6}$$

Which can be rewritten in a compact form by grouping the spatial terms:

$$\frac{\partial \rho}{\partial t} = -\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho v) \tag{1.7}$$

In spherical coordinates, the term on the right-hand side represents the opposite of the divergence of the mass flux density vector ($\rho \mathbf{v}$). We have thus rigorously obtained the **continuity equation for mass** in the Eulerian description:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \tag{1.8}$$

1.1.2 Lagrangian Description

The **Lagrangian description** adopts a different set of independent variables. Instead of position, the mass m contained within a given shell is chosen as the variable. The independent variables are therefore:

- m: mass.

- t: time.

In this formalism, all other quantities become functions of m and t. We will therefore have:

- The density: $\rho = \rho(m, t)$
- The position (the radius of the shell): $r = r(m, t)$

The Lagrangian approach is subject to precise **boundary conditions**:

- For $m = 0$, the position corresponds to the center of the star: $r = 0$.
- For $m = M$ (where M is the total mass of the star), the position corresponds to the stellar surface: $r = R$.

1.1.3 Conceptual Comparison: Euler vs Lagrange

Since we apply these descriptions inside stars, we are studying the behavior of the various shells of stellar matter, but the concepts remain valid for general hydrodynamics. The fundamental difference lies in the observer's point of view:

- **Eulerian Approach:** Attention is fixed on a **fixed point in space** (a specific distance r from the stellar center). The observer watches the flow of the fluid through that point over time, verifying whether the gas is moving inward or outward at that exact coordinate.
  ‘‘‘
- **Lagrangian Approach:** Attention is fixed on a **given element of plasma** (in our one-dimensional case, a shell of mass dm). The observer follows the motion and evolution of that specific matter element over time, verifying whether the shell collapses inward or expands outward.
  ‘‘‘

1.2 Correspondence and Change of Variables between Eulerian and Lagrangian Descriptions

There is a precise spatial and mass correspondence between the two descriptions. In particular, we define the boundaries of the stellar system:

- At the **center of the star**: we have $r = 0$, to which an enclosed mass $m = 0$ corresponds.
- At the **surface of the star**: we have $r = R$ (where R is the total radius of the star), to which $m = M$ corresponds (where M is the total mass of the star).

When passing from the Eulerian to the Lagrangian description, we operate a change of independent variables, going from (r, t) to (m, t). It is essential to keep in mind the chain rules for partial derivatives during this transition.

For example, a partial derivative with respect to mass m is expressed as:

$$\frac{\partial}{\partial m} = \frac{\partial r}{\partial m}\frac{\partial}{\partial r} \tag{1.9}$$

Similarly, for the partial derivative with respect to time t (evaluated at constant mass, i.e., in the Lagrangian description), the chain rule is applied taking into account the dependence of r on time. We obtain:

$$\left(\frac{\partial}{\partial t}\right)_m = \left(\frac{\partial r}{\partial t}\right)_m \frac{\partial}{\partial r} + \left(\frac{\partial}{\partial t}\right)_r \tag{1.10}$$

Where the subscript indicates the variable kept constant during the differentiation operation. When changing variables, it is mandatory to consider how the partial derivatives in one description are connected to those of the other.

1.2.1 Invariance of the Total Mass and the Mass Coordinate

Let us now answer a question raised during the lecture: *in the Lagrangian description, do the mass m or the total mass M change over time?*

The answer is no. The total mass M **does not change over time**. In this context, we are not treating the star formation process, but rather considering an already formed object, endowed with its established initial mass. In the study of **stellar structure** (which we will tackle before stellar evolution), the total mass is fixed and represents a boundary condition of our problem.

Likewise, the "small" mass m (the mass coordinate) does not change over time. In the Lagrangian description, the mass m is assumed as an independent variable and, as such, it is fixed. What is observed is the **position** where the mass shell identified by m is located after a certain time interval (i.e., whether the shell moves outward or inward). Partial derivatives serve precisely to study the variation of quantities while keeping one of the two independent variables fixed.

1.2.2 Transformation Operator (Eulerian-Lagrangian Switch)

Let us return to the expression for the infinitesimal mass of the shell under static conditions:

$$dm = 4\pi r^2 \rho, dr \tag{1.11}$$

We now introduce a mathematical procedure that will allow us to move nimbly from one description to the other. Let us calculate the partial derivative of m with respect to itself, which trivially must equal 1:

$$\frac{\partial m}{\partial m} = 1 \tag{1.12}$$

Exploiting the differential definition just recalled, we can express this derivative as:

$$\frac{\partial m}{\partial m} = 4\pi r^2 \rho \frac{\partial r}{\partial m} = 1 \tag{1.13}$$

This mathematical trick allows us to define a conversion operator. From this relation, we deduce that the partial derivative of *any* quantity with respect to m can be rewritten in terms of the derivative with respect to r:

$$\frac{\partial}{\partial m} = \frac{1}{4\pi r^2 \rho} \frac{\partial}{\partial r} \tag{1.14}$$

Although the Lagrangian approach may initially seem less intuitive than the Eulerian one (since instead of fixing a position in space, one "tracks" the element of volume and mass of the plasma in its displacements), we will work predominantly with the Lagrangian description. The reason lies in the fact that, by exploiting this simple relationship, the Lagrangian approach enormously simplifies many problems of stellar structure.

1.3 Physical Complications in the Stellar Model

We are building the basic tools to tackle stellar structure problems. However, compared to the physics studied previously, we will have to introduce further levels of complexity before getting to the heart of the subject:

1. **Dynamic motions of matter:** Even though we limit ourselves to a one-dimensional model, we cannot treat matter as purely static. We must consider cases in which the star expands or contracts rapidly, operating on a dynamic timescale.

2. **Generalized equation of state:** Stellar matter is not always comparable to an ideal gas. At very high densities, the effects of **quantum mechanics** come into play and the gas becomes **degenerate**. We cannot afford to use disjoint formulations (one for the ideal gas and one for the degenerate gas); we need a **generalized thermodynamics** that allows for a smooth transition between the two regimes, allowing the gas to enter or exit the state of degeneracy.

In summary, the three fundamental pillars on which we will work will be: **gravity**, **thermodynamics** and **quantum mechanics**.

1.4 Treatment of Gravity

1.4.1 Limits of the Newtonian Approach

In this course we will not cover **General Relativity**. Relativistic effects become dominant when studying **compact objects** such as neutron stars or black holes. Although we will conceptually see how a neutron star forms, we will not derive the equations of state or the structure working in the formalism of general relativity (which is the subject of dedicated academic courses). Whenever a relativistic effect becomes relevant, we will simply limit ourselves to mentioning it, maintaining our mathematical framework in the classical limit so as not to overly burden the treatment.

1.4.2 Gravitational Formalism and Potential

We will use the classical expression for the local acceleration of gravity, which we will call g. In the Eulerian description (where a mass shell m is located at position r), the gravitational acceleration experienced by the shell is:

$$g = \frac{Gm}{r^2} \tag{1.15}$$

where G is the universal gravitational constant. Many equations are simplified by the fact that g depends directly on the internal mass m.

Gravity, in our context, is governed by the **Poisson equation**. Denoting the gravitational potential with ϕ, the equation takes the form:

$$\nabla^2\phi = 4\pi G\rho \tag{1.16}$$

Considering spherical geometry in one spatial dimension, the Laplacian operator ∇^2 is expressed as:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) \tag{1.17}$$

We also know that the acceleration of gravity is linked to the gradient of the gravitational potential. In our one-dimensional spatial case, we can write:

$$\frac{\partial\phi}{\partial r} = \frac{Gm}{r^2} \tag{1.18}$$

To derive the gravitational potential $\phi(r)$ in a spherically symmetric star, we integrate the expression from the center of the star up to the radius r:

$$\phi(r) = \int_0^r \frac{Gm}{(r')^2}, dr' + \text{constant} \tag{1.19}$$

This integration constant is generally fixed by imposing the asymptotic boundary condition:

$$\lim_{r\to\infty}\phi(r) = 0 \tag{1.20}$$

Trend of the Potential: By analyzing the general curve of the gravitational potential ϕ for a spherical star, we observe that at the center ($r = 0$) the first derivative is zero; the curve thus starts with zero slope. As one moves outward, with the increase of the mass m enclosed in the shell, the potential grows (assuming less negative values), to then tend asymptotically to zero at infinity. The slope of this curve ($\partial\phi/\partial r$) defines at every point the magnitude of gravity g, which is a vector constantly directed towards the center of the star.

1.5 Properties of Partial Derivatives and Constancy of Time

Resuming the discussion introduced previously, it is fundamental to clarify an essential mathematical concept: when we perform a partial derivative with respect to mass m, by the very definition of partial derivative, all other independent variables (in this case, time t) are kept fixed. Therefore, in this context, time is to be considered constant.

1.6 Hydrostatic Equilibrium (HE)

A fundamental concept in the study of stellar structure is **hydrostatic equilibrium** (often abbreviated as **HE**). The empirical fact that the Sun neither expands nor contracts visibly is the clear proof that hydrostatic equilibrium within a star works extremely effectively.

In hydrodynamics, the condition of hydrostatic equilibrium is equivalent to the **conservation of momentum** equation (or *momentum conservation*).

1.6.1 Eulerian Description of Hydrostatic Equilibrium

Let us consider a spherical shell of infinitesimal thickness dr. The mass contained within this shell is subject to the inward-directed gravitational acceleration. For the star to be in equilibrium, this gravitational force must be perfectly balanced by the pressure difference (i.e., the **pressure gradient**) between the inner and outer part of the shell.

In mathematical terms, the gravitational acceleration multiplied by the density (ρg) is balanced by the pressure gradient. Substituting the expression for the acceleration of gravity $g = \frac{Gm}{r^2}$, we obtain the equation of hydrostatic equilibrium in the **Eulerian description**:

$$\frac{\partial P}{\partial r} = -\frac{Gm}{r^2}\rho \tag{1.21}$$

1.6.2 Lagrangian Description of Hydrostatic Equilibrium

Let us now see how this equation transforms when passing to the **Lagrangian description**. To do so, we apply the transformation rule for the change of variables derived previously, based on the mass continuity equation:

$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho} \tag{1.22}$$

Multiplying the Eulerian equation by this factor, we obtain the expression of the pressure gradient with respect to the mass coordinate m:

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} \tag{1.23}$$

Although at first glance this equation might seem more complex due to the power of r in the denominator, in reality it is considerably simpler. Indeed, we note two crucial differences compared to the Eulerian case:

- The density ρ no longer appears explicitly in the equation.
- The radial dependence goes from proportional to $\frac{1}{r^2}$ to proportional to $\frac{1}{r^4}$.

1.6.3 Stability and Dynamical Timescales

Hydrostatic equilibrium is the condition in which stars find themselves for the vast majority of their lives. It is an equilibrium that is very difficult to violate permanently.

If a star is partially perturbed, it will restore hydrostatic equilibrium in an extremely short time interval, known as the **dynamical timescale** (or hydrodynamic timescale). For example, if we were to perturb the entire structure of the Sun, it would return to hydrostatic equilibrium in just half an hour. This restoring mechanism is extremely powerful; therefore, to significantly violate hydrostatic equilibrium, the initial perturbation must be exceptionally severe and violent. Even in the presence of mild internal perturbations, the equation of hydrostatic equilibrium remains valid except for minuscule percentage violations.

1.7 Purely Mechanical Models of Stellar Structure

Thanks to the condition of hydrostatic equilibrium alone, we are already able to derive some first simple solutions to the problem of stellar structure. We will call these elementary solutions "whiteboard solutions" (*whiteboard simple solutions*). They are based exclusively on the two equations introduced so far, preferably using the Lagrangian description as it is mathematically easier:

1. **Mass continuity:**
$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho} \tag{1.24}$$
“‘

2. **Momentum continuity (Hydrostatic Equilibrium):**
$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} \tag{1.25}$$
“‘

1.7.1 The Problem of System Closure

The system just described consists of only two equations but presents **three unknowns** (the radius r, the density ρ and the pressure P, all as functions of the mass m). To close the problem and obtain an analytical solution, we need a third equation: the **equation of state**.

If we were to use the equation of state of an ideal gas, an additional thermodynamic variable would be introduced, namely the **temperature** (T). Recall that the ideal gas equation of state can be written as:

$$\rho = \frac{\mu}{\mathcal{R}} \frac{P}{T} \tag{1.26}$$

where μ is the **mean molecular weight** and $\mathcal{R}$ is the gas constant. In this generic scenario (where ρ is a function of both P and T), the system would not be closed and would require an additional equation (related to energy transport and generation) to be solved.

1.7.2 Barotropic Gases

However, there are specific physical situations in which the equation of state does not introduce a third variable. These cases occur when we are dealing with **barotropic gases**, i.e., fluids in which the density depends *exclusively* on the pressure: $\rho = \rho(P)$. In such circumstances, a third unknown is not needed and the system formed by mass continuity and hydrostatic equilibrium can be solved autonomously, giving rise to what we call **purely mechanical models**.

Some notable examples of barotropic conditions or gases include:

- **Homogeneous sphere:** the density is assumed constant throughout the star ($\rho = \text{constant}$). This acts directly as the third equation.

- **Isothermal sphere:** a gas kept at a constant temperature throughout its structure.
- **Completely degenerate gas:** the quantum degeneracy pressure depends only on the density (and not on the temperature). This approximation perfectly applies to compact objects such as **white dwarfs** and, in a simplified form, to degenerate matter inside **neutron stars**.
- **Polytropic models:** mathematical abstractions in which the relationship between pressure and density takes the form $P \propto \rho^{\gamma}$. These models prove to be extremely useful; historically, they were used by Subrahmanyan Chandrasekhar to analytically calculate the famous **Chandrasekhar limit mass** for degenerate objects.

1.8 Applications of Hydrostatic Equilibrium: Order of Magnitude Estimates

Based on the hydrostatic equilibrium equation introduced so far, it is possible to derive the estimated value of the pressure and temperature at the center of a star. We are not interested in calculating the exact value at this stage, but rather in determining the order of magnitude, i.e., how we expect pressure and temperature to depend on the mass and radius of the star. This analysis of scaling relations is a very common practice in astrophysics.

1.8.1 Estimate of the Central Pressure

Let us write the hydrostatic equilibrium equation in its Lagrangian form:

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} \tag{1.27}$$

To derive the order of magnitude, we approximate the derivative on the left-hand side as the incremental ratio between the pressure difference (between the surface and the center) and the mass of the star:

$$\frac{\partial P}{\partial m} \approx \frac{P_{sup} - P_c}{M} \tag{1.28}$$

where P_{sup} is the surface pressure and P_c is the central pressure.

To evaluate the right-hand side of the equation, we perform an average by evaluating the variables at half the stellar mass ($m = M/2$) and half the radius ($r = R/2$). The advantage of using the Lagrangian form in this calculation lies in the fact that the density does not explicitly appear in the equation.

By substituting these approximations, we obtain:

$$\frac{P_{sup} - P_c}{M} \approx -\frac{G\left(\frac{M}{2}\right)}{4\pi\left(\frac{R}{2}\right)^4} = -\frac{\frac{GM}{2}}{4\pi\frac{R^4}{16}} = -\frac{2GM}{\pi R^4} \tag{1.29}$$

Since the surface pressure is completely negligible compared to the central pressure ($P_{sup} \ll P_c$), we can eliminate P_{sup} and the negative signs, obtaining for the central pressure:

$$\frac{P_c}{M} \approx \frac{2GM}{\pi R^4} \implies P_c \approx \frac{2GM^2}{\pi R^4} \tag{1.30}$$

Although it is an approximation, this formula provides us with crucial information: the central pressure scales proportionally to the square of the mass divided by the radius to the fourth power. This is the fundamental scaling relation for pressure, which remains valid in general:

$$P_c \propto \frac{M^2}{R^4} \tag{1.31}$$

Numerical example for the Sun:
By inserting the solar parameters into this approximate formula, we obtain a value for the central pressure on the order of:

$$P_c \approx 7 \times 10^{15}, \text{dynes/cm}^2 \tag{1.32}$$

The true value, calculated using advanced computational models, is about 2.4×10^{17} dynes/cm^2. Considering the simplicity of the approximation, we are not too far from the correct order of magnitude.

1.8.2 Estimate of the Central Temperature

To calculate the central temperature, the hydrostatic equilibrium equation is not sufficient; we must resort to an equation of state. We approximate the behavior of stellar matter using the ideal gas equation of state, valid throughout the structure:

$$P = \frac{\rho \mathcal{R} T}{\mu} \implies T = \frac{\mu}{\mathcal{R}} \frac{P}{\rho} \tag{1.33}$$

where μ is the mean molecular weight and $\mathcal{R}$ is the gas constant. Evaluating the equation at the center of the star, the central temperature T_c will be:

$$T_c = \frac{\mu}{\mathcal{R}} \frac{P_c}{\rho_c} \tag{1.34}$$

We now introduce the **mean density** $\bar{\rho}$, defined as the ratio between the total mass and the volume of the star:

$$\bar{\rho} = \frac{M}{\frac{4}{3}\pi R^3} \tag{1.35}$$

To manipulate the equation, we multiply and divide by the mean density $\bar{\rho}$:

$$T_c = \frac{\mu}{\mathcal{R}} P_c \left(\frac{1}{\bar{\rho}}\right) \left(\frac{\bar{\rho}}{\rho_c}\right) \tag{1.36}$$

Substituting the previously found expressions for P_c and $\bar{\rho}$ into the equation, we obtain:

$$T_c = \frac{\mu}{\mathcal{R}} \left(\frac{2GM^2}{\pi R^4}\right) \left(\frac{\frac{4}{3}\pi R^3}{M}\right) \frac{\bar{\rho}}{\rho_c} \tag{1.37}$$

Simplifying the mathematical terms (such as the radius, mass and π), the relation becomes:

$$T_c = \frac{8}{3} \frac{\mu}{\mathcal{R}} \frac{GM}{R} \frac{\bar{\rho}}{\rho_c} \tag{1.38}$$

Why did we isolate the ratio $\bar{\rho}/\rho_c$? Since we do not want to make specific assumptions about the density profile, we can exploit the physical certainty that the mean density is definitely lower than the central density, namely:

$$\frac{\bar{\rho}}{\rho_c} < 1 \tag{1.39}$$

This allows us to define at least an upper limit for the central temperature:

$$T_c \lesssim \frac{8}{3}\frac{\mu}{\mathcal{R}}\frac{GM}{R} \tag{1.40}$$

This density fraction typically assumes values between 0.01 and 0.03.

The fundamental conclusion of this derivation, even simpler than the one concerning pressure, is the **scaling relation for temperature**: the central temperature increases with the mass of the star and decreases with the radius.

$$T_c \propto \frac{M}{R} \tag{1.41}$$

Numerical example for the Sun:

Assuming a mean molecular weight $\mu \approx 0.5$ for the Sun, the limit for the central temperature calculated in this way turns out to be:

$$T_c \lesssim 3 \times 10^7, \mathrm{K} \tag{1.42}$$

This limit is consistent with the true value of the solar central temperature, which stands around 1.6×10^7 K.

These scaling laws (known as "back-of-the-envelope calculations") must be memorized, as they will be widely used in future lectures. The physical motivation is direct: in order for hydrostatic equilibrium to be guaranteed, the pressure must assume a value proportional to M^2/R^4, which, applying the ideal gas law, dictates that the temperature varies proportionally to M/R.

1.9 Stellar Dynamics: Equations of Motion and Timescales

Compared to idealized situations, astrophysics presents complications as stellar structures, in particular evolutionary phases, can vary quite rapidly. Consider, for example, a supernova explosion, an event in which hydrostatic equilibrium is violently violated. To address these scenarios in general terms, it is necessary to introduce the dynamical equations of motion.

1.9.1 Derivation of the Equation of Motion

Let us resume the problem of force balancing. To evaluate the motion of a spherical shell of mass dm, we must calculate the sum of the acting forces per unit area:

1. **Force due to the pressure gradient (per unit area):**
   Exploiting the mass coordinate, this force takes the form: $-\frac{\partial P}{\partial m}dm$.
   ‘‘‘

2. **Gravitational force (per unit area):**
   It is obtained by multiplying the acceleration of gravity (g) by the mass of the shell, and then dividing by the surface area of the shell itself ($4\pi r^2$):

$$-g\left(\frac{dm}{4\pi r^2}\right) = -\frac{Gm}{r^2}\frac{dm}{4\pi r^2} = -\frac{Gm}{4\pi r^4}dm \tag{1.43}$$

“‘

If the system is not in hydrostatic equilibrium, the sum of these two forces does not cancel out, but causes a radial acceleration. According to Newton's second law of motion, the acceleration is the second derivative of the radius with respect to time ($\frac{\partial^2 r}{\partial t^2}$). In terms of force per unit area, we will have:

$$\left(\frac{dm}{4\pi r^2}\right)\frac{\partial^2 r}{\partial t^2} = -\frac{\partial P}{\partial m}dm - \frac{Gm}{4\pi r^4}dm \tag{1.44}$$

Dividing the entire equation by dm, we obtain the general equation of motion of the stellar plasma:

$$\frac{1}{4\pi r^2}\frac{\partial^2 r}{\partial t^2} = -\frac{\partial P}{\partial m} - \frac{Gm}{4\pi r^4} \tag{1.45}$$

Analyzing the signs on the right-hand side: since the pressure decreases moving outward, the derivative $\frac{\partial P}{\partial m}$ is intrinsically negative. Consequently, the term $-\frac{\partial P}{\partial m}$ turns out to be positive, and correctly represents a repulsive force directed toward the outside of the star. The gravitational term remains negative, indicating its attractive nature toward the center.

1.9.2 Gravitational Collapse Timescale (Free-fall Timescale)

The dynamical equation proves extremely useful for defining the timescales of stellar phenomena, which allow establishing in advance which terms of the equation can be neglected in complex physical situations. Given the large variations of orders of magnitude in astrophysics, this simplification is often indispensable.

We can define the timescale by introducing extreme situations:

- Evaluating gravitational collapse in the total absence of pressure.
- Evaluating the explosion by eliminating the force of gravity.

Let us consider the first case. Imagine an unrealistic situation in which we completely cancel the pressure forces: the term $-\frac{\partial P}{\partial m}$ becomes negligible. The star collapses solely under its own weight.

The acceleration can be estimated in absolute value through dimensional analysis, as the ratio between a distance (the stellar radius R) and the square of a time (which we will define as the **free-fall timescale** or τ_{ff}):

$$\frac{\partial^2 r}{\partial t^2} \approx \frac{R}{\tau_{ff}^2} \tag{1.46}$$

Inserting this dimensional approximation into the equation of motion where the pressure has been set to zero, and referring to the entire stellar volume, we obtain:

$$\frac{1}{4\pi R^2}\frac{R}{\tau_{ff}^2} \approx \frac{GM}{4\pi R^4} \tag{1.47}$$

Simplifying the constant terms on both sides, we find the link between kinematic acceleration and acceleration of gravity g:

$$\frac{R}{\tau_{ff}^2} \approx \frac{GM}{R^2} = g \tag{1.48}$$

Finally, isolating the free-fall timescale τ_{ff}, we get:

$$\tau_{ff} \approx \sqrt{\frac{R}{g}} \approx \sqrt{\frac{R^3}{GM}} \tag{1.49}$$

In the next lecture, we will apply a similar procedure to derive the timescale of the thermal explosion.

Chapter 2

Lecture 2 - 07-10-2026

2.1 Introduction: Considerations on the Course Language

Since we are all Italian, we will conduct the lectures in Italian, as it would be superfluous to adopt English in this context. However, should an international student (for example, an Indian boy) arrive in the classroom, I would immediately and without any problem switch to delivering the lecture in English.

2.2 Timescales in Stellar Evolution

In the previous lecture we analyzed what the failure to satisfy hydrodynamic equilibrium would entail (an eventuality that, in practice, is almost completely verified in the vast majority of cases). The star, in the absence of equilibrium, would face two limiting scenarios:

- **Collapse toward the center**: in the event that the contribution of pressure is lacking.
- **Expulsion of the layers (explosion)**: in the event that the effect of gravity is lacking.

Although these are physically realistic situations, they only occur in highly specific conditions. A star, in fact, does not collapse instantaneously onto its own center, since it is not possible to sharply and suddenly cancel (we do not "switch to zero") either pressure or gravity.

However, performing this ideal exercise is fundamental to define **timescales**. Such parameters will prove extremely useful when we go on to complicate the equations, allowing us to evaluate which terms are dominant and relevant over specific time intervals. This dimensional analysis is vital not only for understanding the physics of the various scenarios, but also as a **numerical approximation**: in computational numerical codes, if an effect is not relevant on a certain timescale, the equations are simplified by means of conditional logic loops (such as *"if [condition] then [action]"*).

2.2.1 The Free-Fall Time

We have already defined the **free-fall time**, linked to gravitational collapse. It is essentially expressed as a function of the radius and gravity, and is proportional to the inverse of the square root of gravity itself:

$$t_{\text{ff}} \approx \sqrt{\frac{R}{g}} \tag{2.1}$$

2.2.2 The Explosion Time

On the other hand, ideally assuming gravity to be canceled, we can define an **explosion timescale** (t_{exp}). In the absence of gravity, only the acceleration term linked to the pressure gradient survives in the balance of forces.

The equation for acceleration, expressed as a function of the mass coordinate m, takes into consideration the force per unit area:

$$a = \frac{d^2r}{dt^2} = -4\pi r^2 \frac{\partial P}{\partial m} \tag{2.2}$$

In this context, we consider r as the distance of the shell of matter from the center of the star. By appropriately dividing and multiplying by the unit area ($4\pi r^2$) during the derivation phase, the force per unit mass is obtained. Switching, for convenience, to an **Eulerian description**, we divide by the density and the relation simplifies into the spatial ratio with respect to the radius:

$$a = -\frac{1}{\rho}\frac{\partial P}{\partial r} \tag{2.3}$$

Reasoning by orders of magnitude, the term on the left side (the acceleration) is dimensionally equivalent to a distance divided by a time squared. Assuming the radius of the star R as the reference distance and the explosion time t_{exp} as the time, we obtain:

$$a \approx \frac{R}{t_{\text{exp}}^2} \tag{2.4}$$

The second side of the equation (the pressure term) can be approximated to the order of magnitude:

$$\frac{1}{\rho}\frac{\partial P}{\partial r} \approx \frac{P}{\rho R} \tag{2.5}$$

Equating the two expressions, we can isolate the explosion time:

$$\frac{R}{t_{\text{exp}}^2} \approx \frac{P}{\rho R} \implies t_{\text{exp}}^2 \approx R^2 \frac{\rho}{P} \tag{2.6}$$

Applying the square root we obtain:

$$t_{\text{exp}} \approx R\sqrt{\frac{\rho}{P}} \tag{2.7}$$

Knowing that the term under the root (inverted with respect to how it was just written, namely $\sqrt{P/\rho}$) represents the order of magnitude of the **speed of sound** (v_s) within the stellar medium, we can rewrite the formula as:

$$t_{\text{exp}} \approx \frac{R}{v_s} \tag{2.8}$$

The physical meaning of this result is very intuitive: the explosion timescale indicates, roughly, the time required for a sound wave to travel from the center of the star to its surface, propagating at the characteristic speed of sound of the plasma. This is obviously a simplification, since in real explosion situations shock waves or hydrodynamic shocks can come into play, but it sets a reference timescale in view of more complex analyses.

2.3 The Importance of Scale Quantities in Physics

I would like to open a parenthesis on this issue of timescales and scale quantities. When I was a student, I admit I could not digest these concepts: I wondered how it was possible to be satisfied with such approximate reasoning, rather than resorting to more rigorous and precise calculations. With time and experience, I realized their immense structural importance.

When physical phenomena compose and overlap, having these dimensional quantities as a reference is extremely useful. Therefore, I invite you not to make the mistake I used to make, which is not to underestimate them. At the same time, try to intimately understand their inherent limitations. It is obvious that a stellar explosion does not proceed rigidly with that exact timescale: as one moves outward, differences in pressure and density gradients come into play. However, this estimate provides us with a crucial parameter to compare the weight of the various mathematical terms in the different phases of stellar evolution.

Without the use of orders of magnitude, no progress is made in this field. Unlike pure mathematics, one of the fundamental peculiarities of physics is precisely the need for the researcher to learn to juggle scale quantities, to consciously establish which phenomenon is dominant in a given circumstance and which can instead be legitimately neglected.

2.4 Local Perturbations and Hydrodynamic Time

Broadening the discussion in a more general way, there are situations in which the stellar interior undergoes a **small local perturbation** compared to the global balance of the star (think for example of a wave, an oscillation or a small shock wave). In these circumstances, the plasma of the affected area will autonomously and rapidly tend to bring itself toward an equilibrium condition consistent with its local physical characteristics.

Hydrostatic equilibrium is an extremely powerful mechanism: if it is perturbed (as if the system received a "kick" and moved from its ideal position), the star acts immediately to restore it. If an event locally perturbs the equilibrium condition, the system will return to a balanced state on a very precise timescale, which we call **hydrodynamic time** (t_{idr}).

Since this adjustment process is characterized by the very rapid alternation of compensatory phases comparable to very small "explosions" and "collapses", at the local level the two timescales calculated previously end up converging. The local hydrodynamic time therefore becomes roughly analogous to both the local free-fall time and the local explosion time.

[Editor's note: At this point in the lecture, a very brief informal interruption

occurred due to the arrival in the classroom of a student completely wet from the rain, having traveled by motorcycle without wearing any raincoat on a Thursday of severe bad weather. After a few good-natured jokes about the need to equip oneself with a raincoat in the future, the scientific explanation continued regularly.]

2.4.1 Estimate of the Global Hydrodynamic Time

If we want to quantify the global hydrodynamic time of the star in terms of order of magnitude by equating it to the free-fall time, we use the overall dimensions (the stellar radius R):

$$t_{\text{idr}} \approx t_{\text{ff}} \approx \frac{R}{g^{1/2}} \tag{2.9}$$

Knowing that the surface acceleration of gravity is $g = \frac{GM}{R^2}$, we can substitute it to obtain:

$$t_{\text{idr}} \approx \frac{R}{\left(\frac{GM}{R^2}\right)^{1/2}} = R \cdot \sqrt{\frac{R^2}{GM}} \tag{2.10}$$

From which the fundamental relation is deduced (which also corresponds, ultimately, to a term of the type $\approx (G\bar{\rho})^{-1/2}$):

$$t_{\text{idr}} \propto \sqrt{\frac{R^3}{M}} \tag{2.11}$$

This equation shows us how this equilibrium restoration time scales in proportion to the cube root of the radius divided by the mass.

2.4.2 Application to Different Stellar Types

Based on the Sun, the hydrodynamic time is on the order of **about half an hour**. Since the quantity globally scales as $\sqrt{R^3/M}$, we can make estimates for very different stellar types, starting from the reference value of the Sun:

- **Red Giant (Red Giant Branch)**: Let us consider a giant star with a mass roughly analogous to the solar one ($M \approx 1M_\odot$), but characterized by an expanded radius, about 100 times the solar radius ($R \approx 100R_\odot$). Applying the proportionality relation $\sqrt{R^3/M}$ (which I leave to you as a verification exercise), the hydrodynamic time necessary to restore equilibrium rises dramatically, passing to an order of magnitude of **tens of days**.

  “‘

- **White Dwarf**: Let us consider conversely a very high-density stellar remnant. A white dwarf possesses a comparable mass (between 0.6 and $0.8M_\odot$, thus approximable to around $1M_\odot$), but inscribed in a radius that is barely a fiftieth of the solar one ($R \approx 1/50R_\odot$). Due to this extremely dense matter, the effects are exacerbated and the adjustment occurs in extraordinarily rapid times. Scaling the values, we will speak in this case of a hydrodynamic time on the order of **45 seconds**.

  “‘

It is therefore evident to note how these dynamic effects change scale drastically depending on the evolutionary phase being studied. Having prepared this conceptual ground in advance is the most effective methodological path to consciously tackle the most sophisticated physical problems. We will encounter these advanced mathematical notions, for example, in the last part of the course, where we will treat **asteroseismology**; however, as already anticipated in the previous lecture, for the sake of completeness we will continue to conduct our derivations always remaining, basically, within the approximation of the spherical case.

2.5 Equations of motion and deviations from spherical symmetry

In the event that a non-symmetric problem must be treated, the equations take on a vector form. Adopting the Eulerian description, the equation of motion is expressed as:

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla P - \rho \nabla \Phi \tag{2.12}$$

The derivative operator $\frac{d}{dt}$ takes the name of **substantial derivative** (or material derivative) and is defined by the following relation:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \tag{2.13}$$

In this context, the Laplacian operator applied in Poisson's equation for gravity is the full Laplacian.

These deviations from spherical symmetry become physically relevant should certain phenomena occur, including:

- A strong rotation of the stellar body;
- Interactions with companion bodies;
- Other gravitational interactions.

In general terms, however, an isolated star assumes a strictly spherical geometry.

2.6 Hydrostatic equilibrium in General Relativity

For the sake of completeness, it is necessary to specify that, inside extremely compact objects where the density reaches orders of magnitude close to nuclear density (such as neutron stars and certain white dwarfs), hydrostatic equilibrium undergoes substantial variations due to the effects predicted by General Relativity.

In this regime, the hydrostatic equilibrium equation assumes the form of the well-known **Tolman-Oppenheimer-Volkoff equation**. *(Marginal historical note: Oppenheimer did not exclusively deal with the development of the atomic bomb, but also provided pioneering contributions to the study of neutron stars).*

The equation presents a main term analogous to the classical one, multiplied by a series of corrective factors dictated by the relativistic metric:

$$\frac{dP}{dr} = -\frac{GM\rho}{r^2}\left(1+\frac{P}{\rho c^2}\right)\left(1+\frac{4\pi r^3 P}{Mc^2}\right)\left(1-\frac{2GM}{rc^2}\right)^{-1} \tag{2.14}$$

Observing the equation, it is evident that a first-order approximation ($1 \times 1 \times 1$) yields exactly the classical non-relativistic form. The corrective terms, which operate at the second order, become necessary because the structure of spacetime is different: indeed, the curvature of spacetime is introduced. Strictly speaking, the derivation of this equation starts from Einstein's field equations and requires the introduction of the metric tensor, the Ricci tensor and the Riemann curvature tensor. This complete proof requires dozens of pages of mathematical calculations and goes beyond the scope of this course.

The implications of this physics, however, are central to the understanding of stellar evolution. It is fundamental to determine the evolutionary paths that lead to the formation of the various stellar remnants. The evolution of a star can indeed culminate in radically different scenarios:

- **Brown dwarf** (brown dwarfs);
- **White dwarfs** of carbon and oxygen;
- **White dwarfs** of helium;
- **Neutron stars**;
- **Black holes**;
- Complete explosions (supernovae) in which the star is totally disrupted: in this case no compact remnant is formed (neither a black hole nor a neutron star), with everything resolving into a transient energetic event.

2.7 The Virial Theorem: a generalizable derivation

Let us now address the **virial theorem**. Although it is a concept already known and applied numerous times, it is appropriate to provide a rigorous proof that lends itself to subsequent generalizations.

Normally, the virial theorem is described for an isolated system that is in a state of equilibrium (or close to equilibrium). For a star, this coincides with the condition of hydrostatic equilibrium. In the majority of standard cases, it is assumed that the star is isolated, and that consequently the pressure at its outer limit is strictly zero ($P = 0$). However, there are scenarios in which these conditions must be applied in the presence of a non-zero external pressure. A classic example is the evolution of the core of a star, which is subjected to the pressure exerted by the overlying envelope. Let us therefore re-examine the theorem by writing relationships that allow for this generalization.

Starting from the hydrostatic equilibrium equation, it is particularly advantageous to adopt the **Lagrangian description** instead of the more common Eulerian form. Let us consider the partial derivative with respect to mass m of the term $4\pi r^3 P$:

$$\frac{\partial}{\partial m}\left(4\pi r^3 P\right) = 4\pi r^3 \frac{\partial P}{\partial m} + 12\pi r^2 \frac{\partial r}{\partial m} P \tag{2.15}$$

We proceed by performing an integration in dm over the entire mass of the structure (from the center, $m = 0$, to the total mass, $m = M$):

$$\int_0^M 4\pi r^3 \frac{\partial P}{\partial m} dm = \left[4\pi r^3 P\right]_0^M - \int_0^M 12\pi r^2 \frac{\partial r}{\partial m} P dm \tag{2.16}$$

Let us analyze the first term on the right side of the equation, evaluated at the integration limits ($\left[4\pi r^3 P\right]_0^M$):

- **At the center ($m = 0$):** Since the radius $r = 0$, the term vanishes regardless of the value of the central pressure.
- **At the surface ($m = M$):** The term takes the value $4\pi R^3 P_s$, where R is the total radius and P_s is the surface pressure.

2.7.1 Isolated star case

In the simplest case of an **isolated star** (like the Sun), the pressure at the surface is completely negligible compared to the enormous pressures of the internal layers, so that in excellent approximation we set $P_s = 0$. It follows that the boundary term completely vanishes:

$$\left[4\pi r^3 P\right]_0^M = 4\pi R^3(0) - 0 = 0 \tag{2.17}$$

2.7.2 General case with external pressure

As anticipated, this cancellation is not always verified. If one analyzes the core of a star wrapped by an envelope, the pressure P_s at the interface between the core and the envelope will have a finite value greater than zero. In this case, the boundary term $4\pi R^3 P_s$ does not disappear and profoundly alters the energy balance of the system. For the purposes of this specific lecture, the term will be set to zero assuming the condition of an isolated star, but this contribution will be "resurrected" and reintegrated in future lectures when non-isolated systems are analyzed.

[Marginal note: the lecture is temporarily suspended due to a power outage of the recording equipment.]

2.8 Integration of the Hydrostatic Equilibrium Equation

We begin the analysis by observing a recurring term in the stellar structure equations: the spatial derivative of the radius as a function of mass, namely $\frac{dr}{dm}$, which corresponds to the well-known expression $\frac{1}{4\pi r^2 \rho}$. Substituting it inside the integrands, it is possible to end up with expressions in which the factor $12\pi r^2$ appears (deriving from the volume differential) multiplied by $\frac{1}{4\pi r^2 \rho}$, which allow simplifying the spatial variables involved.

To prove the fundamental energetic relations, we start from the **hydrostatic equilibrium** equation and proceed by integrating it over the entire star. We follow the following steps:

1. **Multiplication:** We multiply both sides of the hydrostatic equilibrium equation by the volume term $4\pi r^3$.
2. **Integration:** We carry out the calculation in detail by making explicit the limits of integration, which go from the center (0) up to the total mass (M). It is a crucial step to explicitly state that we are integrating over the entire mass of the body.

The equation formally becomes:

$$\int_0^M 4\pi r^3 \frac{dP}{dm}, dm = -\int_0^M \frac{Gm}{r}, dm \tag{2.18}$$

Solving the first integral (on the left-hand side) by parts and exploiting the relations seen previously for the differentials, the explicit calculations yield a negative multiplicative factor equal to -3 (deriving from the simplification $\frac{12}{4}$). Simultaneously, the terms containing πr^2 simplify numerator with denominator, leaving inside the integral exclusively the density ρ in the denominator and the pressure P in the numerator. We thus obtain for the left-hand side:

$$-3\int_0^M \frac{P}{\rho}, dm \tag{2.19}$$

For formal convenience and to remove superfluous negative quantities, we perform a change of sign to both sides of the overall equation, thus obtaining positive terms on both sides.

2.9 Definition of Energies and Virial Theorem

2.9.1 Gravitational Energy (E_G)

Let us analyze the right part of the equation. The integrand $\frac{Gm}{r}$ represents the gravitational potential; integrating it over the whole star yields the total gravitational energy. Using the equality symbol by definition ($\equiv$) — a rigorous notation typical of mathematical physics (as taught by the late Professor Anile) — we formally define the **gravitational energy** E_G of the star:

$$E_G \equiv -\int_0^M \frac{Gm}{r}, dm \tag{2.20}$$

The minus sign is intrinsically justified by the fact that we are considering a **bound system**, which, by definition, possesses a negative total energy.

2.9.2 Internal Energy (E_I) and the Pressure/Density Term

The other term of the equation, present on the left, involves the integral of the ratio $\frac{P}{\rho}$. From a physical point of view, this quantity is intimately linked to the square of the speed of sound in the medium. Resorting to the ideal gas equation

of state, we can express it as a function of the specific gas constant and the temperature T:

$$\frac{P}{\rho} = \frac{R}{\mu}T \tag{2.21}$$

From classical thermodynamics, we recall that the specific constant $\frac{R}{\mu}$ can be written as the difference between the specific heat at constant pressure (c_P) and the specific heat at constant volume (c_V):

$$\frac{R}{\mu} = c_P - c_V \tag{2.22}$$

Factoring out c_V, this expression transforms into $c_V(\gamma - 1)$, where $\gamma = \frac{c_P}{c_V}$ represents the **adiabatic constant** (or adiabatic index) of the gas. Substituting into the temperature formula we obtain:

$$\frac{P}{\rho} = c_V(\gamma - 1)T \tag{2.23}$$

In the most common case in basic stellar astrophysics, we treat a **monatomic ideal gas**. Recalling the degrees of freedom of such systems, for a monatomic gas we have $\gamma = \frac{5}{3}$. It follows that the factor $(\gamma - 1)$ is equal to $\frac{2}{3}$. Substituting this value, it is convenient to write:

$$\frac{P}{\rho} = \frac{2}{3}c_V T \tag{2.24}$$

This formalism is particularly advantageous because the product $c_V T$ corresponds exactly to the **internal energy per unit mass**, which we denote with the lowercase letter u:

$$u = c_V T \quad \Longrightarrow \quad \frac{P}{\rho} = \frac{2}{3}u \tag{2.25}$$

Thanks to this intensive relation, we can define the **total internal energy** of the star, denoted by E_I, as the integral of the specific internal energy u over the entire mass of the stellar body:

$$E_I \equiv \int_0^M u, dm \tag{2.26}$$

2.9.3 The Virial Theorem for the Monatomic Ideal Gas

Returning to the integrated hydrostatic equilibrium equation and substituting the newly defined macroscopic quantities, keeping in mind the original multiplicative factor 3 and the thermodynamic coefficient $\frac{2}{3}$, the final relation equates:

- On the left: twice the internal energy ($2E_I$).
- On the right: the gravitational energy changed in sign ($-E_G$).

This formulation constitutes the famous **Virial Theorem**:

$$2E_I = -E_G \quad \Longrightarrow \quad 2E_I + E_G = 0 \tag{2.27}$$

It can therefore be stated that the gravitational energy is equal to minus twice the internal energy of the whole star. It is fundamental to remember that this specific form strictly holds if the medium behaves as a monatomic ideal gas.

2.10 The Generalized Virial Theorem

Since the study of stellar structure requires considering all possible stars and their evolutionary cycles, we will face media that deviate from the monatomic ideal gas model. Molecular mixtures may be present, or the gas could undergo a change of state. It is therefore necessary to extend this treatment to accommodate **any equation of state**.

In the classical theorem just proven, the peculiarity of the monatomic gas is confined uniquely in the coefficient 2 present in the expression $2E_I$, a value derived directly from the assumption that $\gamma = \frac{5}{3}$. The overall balance between internal and gravitational energy always remains valid; simply that numerical coefficient will change depending on the γ of the gas in question and the underlying equation of state.

To generalize, let us characterize an arbitrary equation of state by defining a dimensionless parameter ζ, such that the following relation holds:

$$\zeta u = 3\frac{P}{\rho} \tag{2.28}$$

Every specific state of stellar matter will therefore be governed by a peculiar value of the parameter ζ. Integrating this expression over the mass, the term intended to multiply the internal energy will become ζ (instead of 2). We can therefore write the **Generalized Virial Theorem**:

$$\zeta E_I + E_G = 0 \tag{2.29}$$

2.10.1 Verification on the ideal gas

We can easily verify the validity of this generalization by applying it again to the case of a generic ideal gas. As previously shown:

$$\frac{P}{\rho} = (\gamma - 1)c_V T = (\gamma - 1)u \tag{2.30}$$

Substituting this equivalence into the definition $\zeta u = 3\frac{P}{\rho}$, we determine that the parameter ζ is explicitly linked to the adiabatic index:

$$\zeta = 3(\gamma - 1) \tag{2.31}$$

Reapplying this result to the specific case of the monatomic gas ($\gamma = \frac{5}{3}$), we obtain with a simple arithmetic calculation:

$$\zeta = 3\left(\frac{5}{3} - 1\right) = 3\left(\frac{2}{3}\right) = 2 \tag{2.32}$$

We thus perfectly recover the factor 2 initially proven.

In conclusion, this thermodynamic generalization of the formulation proves to be preparatory and indispensable for the purposes of future developments: in the next phases of the course we will have to adapt the equation of state into even more extended forms, so as to be able to apply the Generalized Virial Theorem also to the physics of **compact objects**.

2.11 Generalization of the Equation of State and the Virial Theorem

The formulation of the virial theorem can be extended to any equation of state, provided the correct proportionality parameter z is identified.

2.11.1 Dependence on the Degrees of Freedom

In the thermodynamic field, the nature of the system and its degrees of freedom determine the ratio between the specific heats at constant pressure and at constant volume:

$$\gamma = \frac{C_p}{C_v} \tag{2.33}$$

The value of γ, in turn, uniquely determines the value of the parameter z. Consequently, to apply the virial theorem to a given system, it is sufficient to calculate the correct ratio between the specific heats of the considered substance.

2.11.2 Extension of the Virial Theorem

The classical formulation of the virial theorem (characterized by the factor $\frac{1}{2}$ commonly encountered in introductory astronomy and cosmology courses) specifically refers to the case of a **monatomic ideal gas**. Here we proceed to its generalization to contexts and fluids having generic equations of state.

2.12 Total Energy of the Configuration

Let us consider a star modeled as a spherical configuration of self-gravitating gas in virial equilibrium. The combination of the virial theorem with the principle of conservation of total energy allows analyzing and solving various physical problems related to stellar structure.

2.12.1 Expression of Total Energy

The total energy E_{tot} of the configuration is given by the sum of the internal energy E_i and the gravitational energy E_g:

$$E_{\text{tot}} = E_i + E_g \tag{2.34}$$

Combining this definition with the generalized virial relation via the parameter z, it is possible to express the total energy in two equivalent forms:

- As a function of the **internal energy** E_i alone:

$$E_{\text{tot}} = (z-1)E_i \tag{2.35}$$

- As a function of the **gravitational energy** E_g alone:

$$E_{\text{tot}} = \frac{z-1}{z}E_g \tag{2.36}$$

2.12.2 Bound Systems and Stability Limit ($\gamma = 4/3$)

In order for the star to constitute a **bound system**, its total energy must be strictly negative:

$$E_{\text{tot}} < 0 \tag{2.37}$$

From the expression of the total energy, the limit condition for stability is directly deduced:

- If $z = 1$, the total energy vanishes: $E_{\text{tot}} = 0$.
- There is therefore a lower limit value for z (and consequently for γ) below which the configuration ceases to be a bound and stable system.

For an ideal gas, the relationship between z and the adiabatic coefficient γ is given by $z = 3(\gamma - 1)$. Setting the limit condition $z = 1$, we obtain:

$$3(\gamma - 1) = 1 \implies \gamma = \frac{4}{3} \tag{2.38}$$

Therefore, a fluid medium characterized by $\gamma = \frac{4}{3}$ is at the limit of stability and presents serious problems for maintaining the equilibrium of the configuration.

2.13 Conservation of Energy and Bolometric Luminosity

To formulate the energy conservation equation for a star, it is necessary to consider that the object emits energy in the form of radiation at the expense of its own total energy.

2.13.1 Energy Balance and Luminosity

From a thermodynamic point of view, the time variation of the total energy of the system must be balanced by the **bolometric luminosity** L, defined as the total energy radiated per unit time at all wavelengths:

$$L = -\frac{dE_{\text{tot}}}{dt} \tag{2.39}$$

By substituting the expressions of the total energy expressed as a function of the parameter z, the energy balance is obtained:

- In terms of internal energy:

$$L = -(z - 1)\frac{dE_i}{dt} \tag{2.40}$$

- In terms of gravitational energy:

$$L = -\frac{z - 1}{z}\frac{dE_g}{dt} \tag{2.41}$$

2.13.2 Monatomic Ideal Gas Case

For a **monatomic ideal gas**, where $\gamma = \frac{5}{3}$, the parameter z assumes the value $z = 2$. By substituting $z = 2$ into the energy balance equations, we obtain:

$$L = \frac{dE_i}{dt} = -\frac{1}{2}\frac{dE_g}{dt} \tag{2.42}$$

2.13.3 Physical Interpretation of Gravitational Contraction

Consider the process of **gravitational contraction** of the star:

1. During the contraction, the time variation of the gravitational energy is negative ($\frac{dE_g}{dt} < 0$), since the star becomes denser and the gravitational energy decreases (becoming more negative).
2. Consequently, the quantity $-\frac{1}{2}\frac{dE_g}{dt}$ turns out to be strictly positive (> 0).
3. This implies that both the luminosity L and the time derivative of the internal energy $\frac{dE_i}{dt}$ are positive quantities (> 0).

This result formalizes the classical formulation of the **virial theorem**: during a gravitational contraction, exactly half of the released gravitational energy is converted into an increase in **internal energy** (causing the heating of the gas), while the other half is radiated outwards in the form of **luminosity**.

2.14 Effects of Gravitational Contraction in the Absence of Nuclear Reactions

2.14.1 Energy Partition and Plasma Heating

Let us consider a star going through a phase of gravitational contraction in the absence of active nuclear reactions within it. Even though it lacks a fusion energy source, the star possesses a non-zero internal energy E_i due to the temperature of the gas and continues to radiate a luminosity L outwards.

Based on the energy balance and the virial theorem, the time variation of the total energy of the system is equal to the emitted luminosity:

$$L = -\frac{dE_{\text{tot}}}{dt} \tag{2.43}$$

The energy released by the gravitational contraction is partitioned equally into two contributions:

- **Half of the variation in gravitational energy** ($-\frac{1}{2}\frac{dE_g}{dt}$) is converted into radiation emitted outwards, determining the luminosity of the star:

$$L = -\frac{1}{2}\frac{dE_g}{dt} \tag{2.44}$$

- **The other half of the variation in gravitational energy** goes to increase the internal energy of the plasma ($\frac{dE_i}{dt}$):

$$\frac{dE_i}{dt} = -\frac{1}{2}\frac{dE_g}{dt} \tag{2.45}$$

2.14.2 Increase of the Internal Temperature

The specific internal energy (per unit mass) of the gas is given by the thermodynamic relation:

$$e_i = c_v T \tag{2.46}$$

where c_v is the specific heat at constant volume and T is the local temperature.

Integrating over the entire mass of the star and assuming that the nature of the gas (and thus the value of c_v) remains unchanged, the increase in internal energy translates into a direct and proportional increment of the plasma temperature T.

2.14.3 Temporal Behavior and Bound State

Since during a contraction the radius of the star decreases, the time variation of the gravitational energy is negative ($\frac{dE_g}{dt} < 0$). Being a gravitationally bound system, the gravitational energy E_g is a negative quantity; consequently, the contraction entails that its absolute value $|E_g|$ increases over time, making the system more energetically bound.

2.15 The Kelvin-Helmholtz Timescale

2.15.1 Definition and Physical Meaning

The **Kelvin-Helmholtz timescale** (τ_{KH}) characterizes the evolutionary phase of a star whose primary energy source is represented exclusively by gravitational contraction, in the absence of nuclear reactions. It typically applies to stars in the **pre-main sequence** phase, namely the phase preceding the ignition of hydrogen burning in the core.

The Kelvin-Helmholtz timescale is defined as the ratio between the absolute value of the available gravitational energy and the radiated luminosity:

$$\tau_{\mathrm{KH}} = \frac{|E_g|}{L} \tag{2.47}$$

Since, by the virial theorem, the gravitational energy and the internal energy are of the same order of magnitude ($|E_g| \sim E_i$), the timescale can be expressed equivalently as:

$$\tau_{\mathrm{KH}} \approx \frac{E_i}{L} \tag{2.48}$$

From a dimensional point of view, dividing an energy by a power (energy per unit time), one correctly obtains a quantity expressed in units of time.

2.15.2 Order of Magnitude Estimate

To formulate an order of magnitude estimate of the gravitational energy E_g, the interaction can be approximated by considering the mean mass and mean radius of the stellar configuration (indicatively assuming $M_{\mathrm{mean}} \sim \frac{M}{2}$ and $R_{\mathrm{mean}} \sim \frac{R}{2}$):

$$|E_g| \sim \frac{GM^2}{2R} \tag{2.49}$$

Substituting this approximation into the definition, the Kelvin-Helmholtz timescale scales proportionally to:

$$\tau_{\mathrm{KH}} \sim \frac{GM^2}{2RL} \tag{2.50}$$

Although during the pre-main sequence contraction both the radius R and the luminosity L vary over time, this formulation provides an excellent estimate of the order of magnitude of the gravitational evolutionary time.

2.15.3 Application to the Sun and Historical Relevance

Evaluating the expression for the current parameters of the Sun ($M = M_\odot$, $R = R_\odot$, $L = L_\odot$), one obtains:

$$\tau_{\mathrm{KH},\odot} \approx 1{,}6 \times 10^7 \text{ years} \tag{2.51}$$

In general, the timescale varies as a function of the factor $\frac{M^2}{RL}$.
This value holds fundamental historical importance:

1. **Classical pre-quantum model**: Before the development of quantum physics and nuclear physics, Kelvin-Helmholtz contraction was believed to be the only known source of solar energy. It was therefore estimated that the total life of the Sun was limited to about 10–20 million years.
2. **Crisis of physical and biological chronology**: This estimate came into sharp contrast with the geological and paleontological evidence that emerged at the end of the 19th century, which demonstrated that the Earth and terrestrial rocks had an age clearly superior to a few tens of millions of years.
3. **Incentive for new physics**: This blatant discrepancy represented one of the main conceptual stimuli that led to the discovery of nuclear fusion reactions and the development of modern stellar astrophysics.

2.16 Concluding Notes

The concepts of virial energy balance and the Kelvin-Helmholtz timescale constitute the essential theoretical foundations for understanding the subsequent phases of stellar evolution and the internal structure of pre-main sequence stars.

2.17 Didactic Indications and Preparatory References

The fundamental concepts covered in the introductory lectures constitute the backbone for the subsequent theoretical developments of stellar physics. These topics, despite being reviewed within preparatory courses (such as *Foundations of Astrophysics* or *Elements of Astronomy and Cosmology*), are presented with a higher level of mathematical formalism, precision, and punctuality. For this reason, an in-depth and timely study of these conceptual bases is recommended, which is indispensable for fluently tackling the more complex contents of the subsequent lectures.

2.18 The Virial Theorem in the Presence of Non-Zero External Pressure

Let us consider the generalization of the virial theorem in the case where the pressure at the outer surface of the stellar structure, denoted by P_0, is not negligible ($P_0 \neq 0$).

In the presence of a non-zero external pressure, the surface term — previously omitted in the simplified derivations — must be explicitly maintained in the integrated balance. The integral equality between the gravitational potential energy and the pressure term takes the following form:

$$\int_0^M \frac{d\Phi}{dm}, dm = 3 \int_0^M \frac{P}{\rho}, dm - 4\pi R^3 P_0 \tag{2.52}$$

where M is the total mass of the star, R is the surface radius, P is the local pressure and ρ is the mass density. In the limiting case where $P_0 = 0$, the boundary term $-4\pi R^3 P_0$ vanishes, leading the expression back to the standard form.

The formulation of the virial theorem modified by the presence of the external pressure P_0 is expressed as:

$$2E_i + E_g = 4\pi R^3 P_0 \tag{2.53}$$

in which E_i represents the total internal energy of the system and E_g the gravitational potential energy.

2.18.1 Dimensional Analysis of the Surface Term

The dimensional consistency of the relation expressed by equation (2) finds direct confirmation in the definition of the physical quantities involved:

- The pressure P_0 represents a force per unit area ($[\mathrm{N/m^2}]$).
- By multiplying the pressure by an area proportional to R^2, one obtains a dimension equivalent to that of a force ([N]).
- A further multiplication by a length R (obtaining the dependence on R^3) defines a work, i.e., a quantity having the physical dimensions of an energy ([J]).

Therefore, the term $4\pi R^3 P_0$ represents an energetic contribution consistent with the terms E_i and E_g.

2.19 Summary of Key Concepts for Stellar Physics and Structure

At the conclusion of the treatment, the fundamental principles and the analytical relations established, essential for the continuation of the study of stellar structure and evolution, are summarized.

2.19.1 Fundamental Equations of Balance and State

- **Mass continuity equation**: defines the spatial mass distribution within the stellar body.
- **Momentum conservation equation**: defines the hydrostatic equilibrium condition of the stellar structure.
- **Role of the equation of state**: depending on the analytical form of the adopted equation of state, it is possible to determine a complete mechanical solution of the structure without the need to resort to further equations related to energy transport and balance.

2.19.2 Scaling Relations (*Scaling Laws*)

The main internal thermodynamic quantities depend on the mass M and the radius R of the stellar body according to the following proportionalities:

- **Internal pressure**: $P \propto \frac{M^2}{R^4}$
- **Internal temperature**: $T \propto \frac{M}{R}$

2.19.3 Characteristic Timescales

Dynamical and evolutionary processes are governed by the following foundational timescales:

- **Free-fall timescale** ($\tau_{\rm ff}$, *free-fall timescale*).
- **Explosion timescale**.
- **Hydrodynamic timescale** ($\tau_{\rm dyn}$).

2.19.4 Virial Theorem and Thermodynamic Implications

- General formulation and application to the case of the **monatomic ideal gas**.
- **Conservation of energy during contraction**: in the absence of nuclear energy sources, the gravitational contraction of a star entails that:
  - Half of the variation in gravitational energy ($\frac{1}{2}|\Delta E_g|$) is converted into luminosity L and radiated outwards.
  - The remaining half goes to increase the internal energy (ΔE_i), causing an increase in the internal temperature of the plasma.
- **Kelvin-Helmholtz timescale** ($\tau_{\rm KH}$): represents the typical evolutionary timescale of stars in the pre-main sequence phase, during which gravitational contraction constitutes the only energy source of the star.

2.20 Organizational Notes and Schedule Management

Regarding the requests to anticipate the schedule of the lectures, we specify the impossibility of modifying the calendar due to institutional teaching overlaps. In particular, the professor is engaged in conducting the course of *Foundations of Astrophysics* on Mondays from 15:00 to 18:00.

Chapter 3

Lesson 3 -

Chapter 4

Lesson 4 -

Chapter 5

Lecture 5 - 20/10/2026

5.1 Conservation of Energy: Global and Local View

Taking up the concepts previously discussed, we focus on the **conservation of energy**, analyzing it from both a global and a local point of view.

Analyzing the star layer by layer (or "shell" by "shell"), the local conservation equation describes how energy is transferred within the structure. As long as there is nuclear energy production, the luminosity increases. When the outermost layers are reached where there is no longer any nuclear burning, the luminosity is conserved and becomes constant. Mathematically, this translates into:

$$\frac{dL}{dm} = 0 \implies L = \text{constant} \tag{5.1}$$

Dividing by $4\pi r^2$, one obtains the corresponding energy flux, which is diluted over the entire spherical surface. In the absence of nuclear production, the overall luminosity is transferred to the surface in a constant manner.

However, one must consider possible contributions deriving from **thermodynamic work**. If the stellar layers are expanding or contracting, it is necessary to take into account the work involving pressure (P) and volume (V), as well as the transfer of energy into internal energy, or, conversely, the use of internal energy to perform work.

The local energy equation can be written in its most general form taking into account all possibilities: nuclear reactions, neutrino losses, thermodynamic work and deviations from ideal gas behavior.

From a global point of view, the luminosity produced is strictly linked to the variation in total energy. This luminosity is composed of both photons and neutrinos. The relationship between global energy and local energy is expressed via the following integrals:

- The integral of the local photon luminosity yields the **total photon luminosity**:

$$L_{photons} = \int dL \tag{5.2}$$

- The integral of the energy loss via neutrinos yields the **neutrino luminosity**:

$$L_\nu = \int \epsilon_\nu, dm \tag{5.3}$$

5.1.1 Gravitational Energy and Internal Energy

The link between the total energy terms and the gravitational energy requires a reference to the **virial theorem**. We have seen that the integral of the local internal energy over the mass yields the variation of global internal energy:

$$E_{int} = \int_0^M u, dm \tag{5.4}$$

As for the **gravitational energy** (E_G), from the virial theorem one derives:

$$E_G = -3 \int \frac{P}{\rho}, dm \tag{5.5}$$

Expressing the time derivative (local gravitational work), one obtains a term proportional to $\frac{GM}{r}\frac{\dot{r}}{r}$, which through the virial relations can be rewritten in terms of density variation:

$$\dot{E} * G = \int *0^M \frac{P}{\rho^2}\dot{\rho}, dm \tag{5.6}$$

The expression for total energy involves kinetic energy, gravitational energy, internal energy and nuclear energy. Assuming **hydrostatic equilibrium** (there are no global gas movements within the star), the macroscopic kinetic energy is zero ($E_{kin} = 0$).

5.2 Stellar Timescales

5.2.1 The Nuclear Timescale

Dealing with nuclear energy, we introduce the **nuclear timescale** (τ_{nuc}), defined in a manner analogous to the other timescales already seen:

$$\tau_{nuc} = \frac{E_{nuc}}{L} \tag{5.7}$$

where E_{nuc} is the available nuclear energy (expressed in ergs) and L is the luminosity (in erg/s).

In the case of the Sun, this calculation provides a nuclear timescale of about 10^{10} years. This result resolved the classic historical problem of the age of the Sun:

- The **Kelvin-Helmholtz** thermal timescale provided an age of barely $\tau_{KH} \approx 10^7$ years.
- The **nuclear** timescale provides $\tau_{nuc} \approx 10^{10}$ years, a value perfectly compatible with geological evidence.

5.2.2 Hierarchy of Timescales

For the phase in which hydrogen or helium is burning (*hydrogen / helium burning*), since these elements are very abundant and their cross sections are not excessively high, the following hierarchy for stellar timescales holds:

$$\tau_{nuc} \gg \tau_{KH} \gg \tau_{hydro} \tag{5.8}$$

For the Sun, we have 10^{10} years for the nuclear, 10^7 years for Kelvin-Helmholtz and about 30 minutes for the **hydrodynamic** timescale (τ_{hydro}), which represents the time required to bring the star back into equilibrium in the event of large-scale perturbations.

5.3 Order of Magnitude Analysis of the Energy Equation

Let us take the local energy conservation equation in its full differential form:

$$\frac{\partial L}{\partial m} = \epsilon_{nuc} - \epsilon_{\nu} + \epsilon_g \tag{5.9}$$

where ϵ_g represents the thermo-gravitational term (a combination of work and internal energy variation). We express each term in orders of magnitude using the timescales just introduced to understand which terms dominate and which can be neglected depending on the phenomenon.

1. **Spatial variation of luminosity:**

$$\frac{\partial L}{\partial m} \sim \frac{L}{M} \tag{5.10}$$

   Multiplying and dividing by the total internal energy E_{int}, we can rewrite it as a function of the Kelvin-Helmholtz timescale:

$$\frac{L}{M} \sim \frac{E_{int}}{M, \tau_{KH}} \tag{5.11}$$

   since $\tau_{KH} \approx \frac{E_{int}}{L} \approx \frac{|E_G|}{L}$.
   ‘‘‘

2. **Production of nuclear energy:**

$$\epsilon_{nuc} \sim \frac{L_{nuc}}{M} \tag{5.12}$$

   Multiplying and dividing by the total nuclear energy E_{nuc}, we obtain:

$$\epsilon_{nuc} \sim \frac{E_{nuc}}{M\,\tau_{nuc}} \tag{5.13}$$

   The variation of L therefore scales with the Kelvin-Helmholtz time and depends on the available internal energy, while the rate ϵ_{nuc} scales with the nuclear time and depends on the available nuclear energy.

3. **Thermo-gravitational term (ϵ_g):** From the first law of thermodynamics (linked to the variation of entropy over time $T\frac{dS}{dt}$), the local term ϵ_g is expressed as:

$$\epsilon_g = -C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho}\frac{\partial P}{\partial t} \tag{5.14}$$

In orders of magnitude, the time derivative introduces the timescale of the *current* phenomenon in progress (τ):

$$-C_P \frac{\partial T}{\partial t} \sim \frac{u}{M\,\tau} \tag{5.15}$$

where u is the internal energy density. Specifying for an **ideal gas** ($\delta = 1$ and $\frac{P}{\rho} = \frac{R}{\mu}T$), one finds that the pressure term also has the same order of magnitude:

$$\frac{\delta}{\rho}\frac{\partial P}{\partial t} \sim \frac{R}{\mu}\frac{T}{\tau} \sim \frac{u}{M\,\tau} \tag{5.16}$$

‘‘

5.4 Applications in Various Evolutionary Scenarios

This approach allows us to enormously simplify the energy equation by ignoring, case by case, the terms that are orders of magnitude lower compared to the others.

5.4.1 Case 1: Evolution on the Main Sequence ($\tau \gg \tau_{KH}$)

When analyzing evolution on a nuclear scale (such as during the Main Sequence), the variation of molecular weight and stellar evolution occur on a very long timescale:

$$\tau = \tau_{nuc} \gg \tau_{KH} \tag{5.17}$$

In this regime, the term $\epsilon_g \propto \frac{1}{\tau_{nuc}}$ becomes negligible compared to the terms $\frac{\partial L}{\partial m} \propto \frac{1}{\tau_{KH}}$. The thermo-gravitational terms (or *gravothermal*, as defined by Kippenhahn) can be completely ignored. The equation reduces to the classic stationary form:

$$\frac{\partial L}{\partial m} \approx \epsilon_{nuc} - \epsilon_\nu \tag{5.18}$$

5.4.2 Case 2: Rapid Evolution, Explosions and Pulsations ($\tau \ll \tau_{KH}$)

Consider instead extremely rapid phenomena, such as a supernova explosion (timescale of days) or the pulsation of a Cepheid. In these cases the timescale of the phenomenon is of the order of the hydrodynamic timescale, neatly lower than that of Kelvin-Helmholtz:

$$\tau \approx \tau_{hydro} \ll \tau_{KH} \tag{5.19}$$

Since $\frac{\partial L}{\partial m} \propto \frac{1}{\tau_{KH}}$, the variation in luminosity is very slow compared to the phenomenon underway, making the term $\frac{\partial L}{\partial m}$ approximately zero compared to the thermo-gravitational terms. The dominant term is precisely ϵ_g, and the equation is simplified by imposing $\epsilon_g \approx 0$ (assuming the absence of strong nuclear reactions in the same hydrodynamic interval), which is equivalent to describing an **adiabatic** evolution ($dS \approx 0$). In physical terms:

$$\epsilon_g = 0 \implies -C_P \frac{\partial T}{\partial t} + \frac{\delta}{\rho} \frac{\partial P}{\partial t} = 0 \tag{5.20}$$

$$C_P \frac{\partial T}{\partial t} \approx \frac{\delta}{\rho} \frac{\partial P}{\partial t} \tag{5.21}$$

In the specific case of a **Cepheid**, the pulsation involves the envelope of the star, which contracts and expands significantly over very short timescales (about one day). Since the time required to dissipate that energy radiatively is dictated by τ_{KH} (much longer), the gaseous layer in motion does not have the physical time to exchange energy with the outside effectively. The variation in internal energy is therefore entirely compensated by the thermodynamic work of the gas, confirming that the expansion and contraction cycle takes place in almost strictly adiabatic conditions.

5.5 Energy Transport Within Stars

The study of internal stellar structure requires understanding how energy is transported from the core toward the outermost layers. The main modes through which this transfer occurs are three:

- **Radiative transport**;
- **Conduction transport**;
- **Convection transport**.

In different physical conditions, different forms of transport will prevail. These modes significantly control the thermodynamic structure of the star. Consequently, the calculation of energy transport joins the other fundamental equations of stellar structure, namely mass conservation, momentum conservation (which translates into hydrostatic equilibrium) and energy conservation.

5.6 Radiative Transport as Diffusion

The problem of radiation transport is, in the general case, extremely complex since it involves in detail the interaction between radiation and matter (as happens, for example, in the study of stellar atmospheres).

However, inside the star it is possible to adopt a very strong approximation. Since the density ρ is extremely high, the **mean free path** (l) of the photons is very small. This allows us to treat radiation transport in stellar interiors with a formal approach analogous to that of **particle diffusion**, where the particles in question are photons.

The mean free path is inversely proportional to the **opacity** of matter (which we will denote by κ) and to the density ρ. Since opacity is expressed as a cross section per unit mass (in $\mathrm{cm^2 g^{-1}}$), the mean free path takes the form:

$$l = \frac{1}{\kappa \rho} \tag{5.22}$$

To give an order of magnitude, inside the Sun the mean free path of photons is about 2 cm. Let us indeed consider typical values for stellar interiors:

- The global average opacity can be of the order of $\kappa \approx 1$ $\mathrm{cm^2 g^{-1}}$.
- If we specifically consider electron *scattering*, the opacity is $\kappa_{es} \approx 0.4$ $\mathrm{cm^2 g^{-1}}$.
- With an average solar density of about $\rho \approx 1.4$ g $\mathrm{cm^{-3}}$, the product $\kappa\rho$ leads to travel distances on the order of a centimeter.

Since the mean free path is infinitesimal compared to the overall dimensions of the star, photons are continuously emitted and reabsorbed over very short distances. This fully justifies the use of a diffusion regime.

5.6.1 The Diffusion Equation for Photons

In a classical diffusion regime, the particle flux $\vec{J}$ is proportional to the density gradient of the particles themselves, mediated by a **diffusion coefficient** D:

$$\vec{J} = -D \nabla n \tag{5.23}$$

In the simple non-relativistic case, the diffusion coefficient is given by:

$$D = \frac{1}{3} v l \tag{5.24}$$

where the factor 1/3 derives from the three spatial degrees of freedom, v is the average particle velocity and l is their mean free path.

Let us now adapt this formalism to photons. Their velocity is equal to the speed of light, $v = c$ (neglecting the variations of the refractive index in the stellar plasma). Instead of the particle number density, we reason in terms of **radiation energy density**, which we will denote by U.

Being in stellar interiors, we are strictly in conditions of **Local Thermodynamic Equilibrium (LTE)**. Under such conditions, radiation behaves like a black body and the photon energy density is proportional to the fourth power of temperature T:

$$U = aT^4 \tag{5.25}$$

where a is the radiation constant.

Given the symmetries of the stellar problem, all physical quantities vary exclusively as a function of the radial coordinate r. The radiative flux $\vec{F}_{rad}$ thus reduces to a scalar directed along the radius. By substituting the terms into the diffusion law, we obtain:

$$F_{rad} = -D \frac{\partial U}{\partial r} = -\left(\frac{1}{3} c \frac{1}{\kappa \rho}\right) \frac{\partial (aT^4)}{\partial r} \tag{5.26}$$

By performing the derivative with respect to the radius, and considering a as a constant, we arrive at the final formulation for the radiative flux:

$$F_{rad} = -\frac{4}{3}\frac{acT^3}{\kappa\rho}\frac{dT}{dr} \tag{5.27}$$

In this way, we have reduced the complex problem of radiation-matter interaction to a simple gradient equation.

5.7 The Temperature Gradient in Stellar Structure

From a formal point of view, the newly derived equation highlights that the photon flux is driven by the **temperature gradient**. We can rewrite it by defining a radiative conductivity coefficient, which we will denote by $\mathcal{K}_{rad}$ to distinguish it from the opacity κ:

$$F_{rad} = -\mathcal{K} * rad\frac{dT}{dr} \tag{5.28}$$

where the coefficient is equal to:

$$\mathcal{K} * rad = \frac{4acT^3}{3\kappa\rho} \tag{5.29}$$

This formulation provides us with an important physical interpretation:

- The radiative flux is **favored** by high temperatures (T^3 in the numerator), since the layer struggles less to transfer energy.
- The flux is **hindered** (slowed down) by opacity κ and density ρ. A denser or more opaque medium provides greater "resistance" to the passage of radiation, reducing the mean free path and making energy transport more difficult.

5.7.1 Explicit Form for Structure Calculations

In stellar structure codes and calculations, the equation is reversed: starting from the generated energy flux, the temperature gradient necessary to dissipate it is calculated. By inverting the formula we obtain:

$$\frac{dT}{dr} = -\frac{3\kappa\rho}{4acT^3}F_{rad} \tag{5.30}$$

However, in astrophysical practice, instead of the flux F_{rad}, the **luminosity** L is used, which represents the total energy radiated through a spherical surface of radius r:

$$F_{rad} = \frac{L}{4\pi r^2} \tag{5.31}$$

By substituting the flux, the spatial temperature gradient becomes:

$$\frac{dT}{dr} = -\frac{3\kappa\rho L}{16\pi acT^3 r^2} \tag{5.32}$$

Often, stellar structure models use the enclosed mass m as the independent variable instead of the radius r. To perform the change of variable, we exploit the mass conservation equation:

$$dm = 4\pi r^2 \rho, dr \implies \frac{dr}{dm} = \frac{1}{4\pi r^2 \rho} \tag{5.33}$$

Applying the chain rule for derivatives, we calculate $\frac{dT}{dm}$:

$$\frac{dT}{dm} = \frac{dT}{dr}\frac{dr}{dm} = \left(-\frac{3\kappa\rho L}{16\pi acT^3 r^2}\right)\left(\frac{1}{4\pi r^2 \rho}\right) \tag{5.34}$$

By simplifying the density ρ and multiplying the terms in the denominator, we obtain the definitive differential equation for the temperature gradient as a function of mass:

$$\frac{dT}{dm} = -\frac{3\kappa L}{64\pi^2 acT^3 r^4} \tag{5.35}$$

In summary, the physics of radiative transport teaches us that, given a certain energy flux in transit outwards, the temperature gradient that is established depends on the characteristics of the layer: if opacity and density increase, the layer opposes greater resistance, and the temperature gradient required to maintain the flux must necessarily increase. If the temperature increases, the resistance decreases and the gradient is reduced.

5.8 Transport by Conduction

In addition to radiation, energy can be transferred via **thermal conduction**. In classical stellar interiors, transport by conduction is considered secondary and often negligible, since the conductivity of the plasma is such that it does not compete with the efficiency of radiative transport.

However, in more accurate physical calculations (and in particular stellar regimes), conduction is included in the equations. The mathematical approach is entirely analogous to the one just seen for photon diffusion: the classical form of the conduction equation applied to the gas particles is used. The conductive flux will be proportional to the temperature gradient via a specific thermal conductivity coefficient:

$$F_{cond} = -\mathcal{K}_{cond}\frac{dT}{dr} \tag{5.36}$$

The two fluxes (radiative and conductive) can then be formally added together in the study of the global energy balance of the layer. The study of internal stellar structure requires understanding how energy is transported from the core toward the outermost layers. The main modes through which this transfer occurs are three:

- **Radiative transport**;
- **Conduction transport**;
- **Convection transport**.

In different physical conditions, different forms of transport will prevail. These modes significantly control the thermodynamic structure of the star. Consequently, the calculation of energy transport joins the other fundamental equations of stellar structure, namely mass conservation, momentum conservation (which translates into hydrostatic equilibrium) and energy conservation.

5.9 Energy Transport: Conduction and Total Opacity

As previously introduced, energy transport via thermal conduction, associated with the motion of gas particles, can be treated analytically by exploiting the classical form of conduction. Analogously to the case of radiative transport, it is possible to define a conductive coefficient $\mathcal{K}_{cond}$ that links the energy flux to the temperature gradient.

In stellar interiors, it is extremely convenient to express the conduction coefficient in a mathematical form completely similar to that of the radiative coefficient. For this purpose, a **conductive opacity** is introduced, denoted by κ_{cond}. The total energy flux, F_{tot}, will be given by the sum of the radiative flux and the conductive flux:

$$F_{tot} = F_{rad} + F_{cond} = -\left(\mathcal{K} * rad + \mathcal{K} * cond\right) \nabla T \tag{5.37}$$

Making the terms explicit as a function of the radiative opacity (κ_{rad}) and the conductive one (κ_{cond}), one obtains:

$$F_{tot} = -\frac{4ac}{3}\frac{T^3}{\rho}\left(\frac{1}{\kappa_{rad}} + \frac{1}{\kappa_{cond}}\right)\nabla T \tag{5.38}$$

5.9.1 The formal invariance of the constant c

A question naturally arises: why does the speed of light c continue to appear in the total flux expression within the factor $\frac{4ac}{3}$, even for the conductive term which concerns massive particles and not photons?

This is a precise choice of formal convenience. The actual speed of the gas particles is absorbed (incorporated) within the definition of the conductive opacity κ_{cond}, which is then appropriately scaled by the speed of light c. This mathematical artifice allows factoring out the term $\frac{4ac}{3}$, maintaining the structure of the differential equation unchanged.

The result of this operation translates into a simple correction of the opacity coefficient. It is sufficient to define a single **total opacity** κ (or corrected opacity) such that:

$$\frac{1}{\kappa} = \frac{1}{\kappa_{rad}} + \frac{1}{\kappa_{cond}} \tag{5.39}$$

In this way, the treatment of energy transport is significantly simplified, since the effect of conduction is entirely absorbed into the definition of the equivalent opacity of the medium.

5.10 The Calculation of Opacities and Stellar Structure Databases

The mathematical simplification just seen finds its practical application in numerical calculation codes. The opacity parameter κ is calculated a priori, since its analytical computation is a problem of atomic physics of enormous complexity.

In stellar interiors, due to the high energies involved, it is necessary to take into account countless atomic transitions, including those of heavy and complex elements such as iron or vanadium. Between the 90s and the early 2000s, the scientific community made an extraordinary computational effort through international programs such as the **Opacity Project** and the **Iron Project**. These projects, exploiting the maximum computational capabilities of the time, calculated all the necessary cross sections to build realistic opacity tables.

The opacity of a stellar mixture strictly depends on:

- Chemical composition of the gas;
- Temperature (T);
- Density (ρ);
- Equation of state of the gas (which takes into account, for example, the electron degeneracy regime that is established at very high densities and not extremely high temperatures).

Today, modern stellar evolution codes do not calculate the opacity run-time, but directly query vast **grid databases**. By calling a specific routine, the code extrapolates the correct κ value, which already includes within it the contribution of conductive transport. These tables are essential for modeling stars in any phase, including advanced evolutionary phases (e.g., AGB stars) where nuclear reactions produce heavy elements that further complicate radiation-matter interaction.

5.11 Radiative Temperature Gradient with respect to Pressure

Let us return to the temperature gradient equation for radiative transport in Lagrangian coordinates (mass m). As previously demonstrated, the physical variables involved are opacity, luminosity, distance from the center r and temperature T.

$$\frac{dT}{dm} = -\frac{3\kappa L}{64\pi^2 acT^3 r^4} \tag{5.40}$$

To study the thermodynamic structure as a function of pressure, we combine this equation with the **hydrostatic equilibrium** equation, which expresses the pressure gradient with respect to mass:

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \tag{5.41}$$

To obtain the temperature gradient with respect to pressure ($\frac{dT}{dP}$), we take the member-by-member ratio between the two differential equations:

$$\frac{dT}{dP} = \frac{\left(\frac{dT}{dm}\right)}{\left(\frac{dP}{dm}\right)} = \frac{\left(-\frac{3\kappa L}{64\pi^2 acT^3 r^4}\right)}{\left(-\frac{Gm}{4\pi r^4}\right)} \tag{5.42}$$

Performing the algebraic simplifications:

- The spatial term r^4 in the denominator cancels out on both sides.
- The π^2 term in the numerator simplifies with the π in the denominator, leaving a π in the denominator in the final result.
- The numerical coefficient is reduced: $\frac{4}{64} = \frac{1}{16}$.

The **radiative temperature gradient** (∇_{rad}) is thus obtained:

$$\frac{dT}{dP} = \frac{3\kappa L}{16\pi acGmT^3} \tag{5.43}$$

This formulation allows expressing the local thermal variation directly in terms of pressure, enclosed mass and luminosity.

5.11.1 Note on the density variable (ρ)

It is important to make an algebraic clarification regarding the absence of the density ρ in the $\frac{dT}{dm}$ equation. During the transition from Eulerian coordinates (radius r) to Lagrangian coordinates (mass m), the differential relation of mass conservation is used: $dm = 4\pi r^2 \rho dr$. Dividing the initial expression by this term, the density ρ (present in the numerator in the equation in dr) exactly simplifies with that introduced by the mass differential. This represents one of the major advantages of the Lagrangian description: density is absorbed by the mass variable, significantly simplifying the system of equations.

Marginal note: Guidance on the Study Plan

In response to a question asked at the end of the lecture regarding the overlaps between the Cosmology (or Elements of Astronomy) course of the bachelor's degree and the programs of the master's degree: it is emphasized that a large part of the introductory topics covered in the bachelor's scope will be taken up and deepened in the various master's courses. The choice of the master's path is not absolutely bound by the optional subjects taken at the bachelor's level. The recommended approach is to select the elective exams based on one's main inclinations and interests; any previous gaps related to non-attended subjects can be easily filled during the master's studies.

Chapter 6

Lesson 3 -